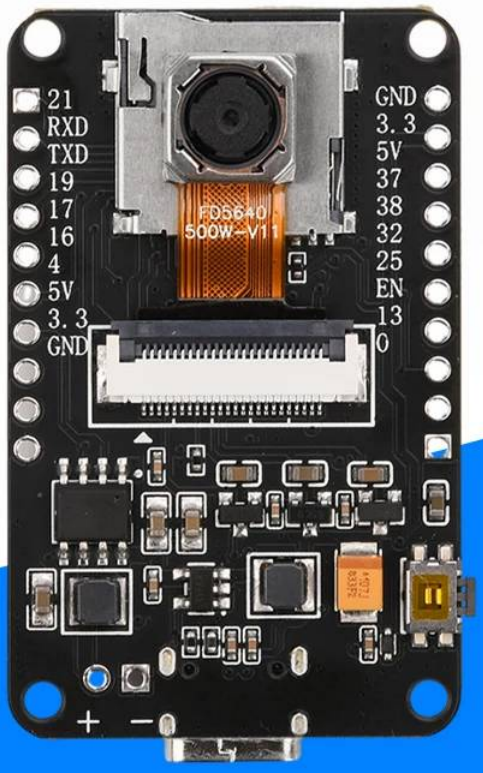
Features: 1. This development board is an upgraded version of the ESP32-CAM development board, which adds PSRAM. 2. The module adopts ESP32 low-power solution, integrates WiFi and Bluetooth, and can operate independently as the smallest system. 3. The module is equipped with a default 0V5640 camera, which, combined with the powerful performance of ESP32, can reach up to 30 FPS in VGA format. 4. The most significant feature of this upgrade is the addition of 8MB (bytes) of external RAM and upgraded TF card slot, which supports users to use up to 32GB of storage cards and can capture higher pixels and store more images, meeting various secondary development needs of users. 5. And after joining PSRAM, it can also support the facial recognition framework ESP-WHO, which is suitable for use in IoT and smart device scenarios, such as the development of facial recognition projects for supermarket lockers. Parameter: MCU: ESP32-D0WDQ6: Frequency adjustable between 80-240MHz, integrated 2.4GHz Wi Fi and Bluetooth dual-mode, 40nm process, internally integrated 520KB RAM and onboard 8MB PSRAM, using SPI protocol communication, with a speed of up to 40M. USB: CH340X: Automatic download circuit, equipped with E8051 core compatible with MCS51 instruction set, with an average instruction speed 8-15 times faster than standard MCS51. ESP32-CAM module uses CH340X to implement automatic download circuit, and with USB, it is easy to write and debug ESP32. CMOS: OV5640 Camera: Provides the complete functionality of a single-chip JXGA (1632x1232) camera and image processor, offering multiple formats of full frame sampling, scaling, or windowing 8-bit/10 bit images through the serial camera control bus (SCCB) interface, with an optimal image distance of 20-250cm. Dual power supply mode: USB and external battery: reserved battery welding points, default does not include battery, users can connect 3.7V lithium-ion battery by themselves. The module is equipped with IP5306 power management chip to detect the battery level/charging status. In battery power supply mode, double-click the reset button to turn off the device, and click the reset button to turn on the device. HY2.0 cable interface: A HY2.0 cable interface is reserved and mounted on the internal I2C bus to enable it Multi functionality can be achieved through I2C extension to meet different application scenarios. Trans flash Card: Supports up to 32GB of storage card Sensor Category: Optical Sensor Working current: 0.15A Working voltage: 5V Size: 48mm * 30mm Module application scenarios: Secondary development of Aolt applications such as monitoring, video, and photography Low cost camera solution for IoT node devices Package include: 1X development board 1X OV5640 Camera 1X radiating rib

WIFI BT SMART MODULE

ESP32-CAM

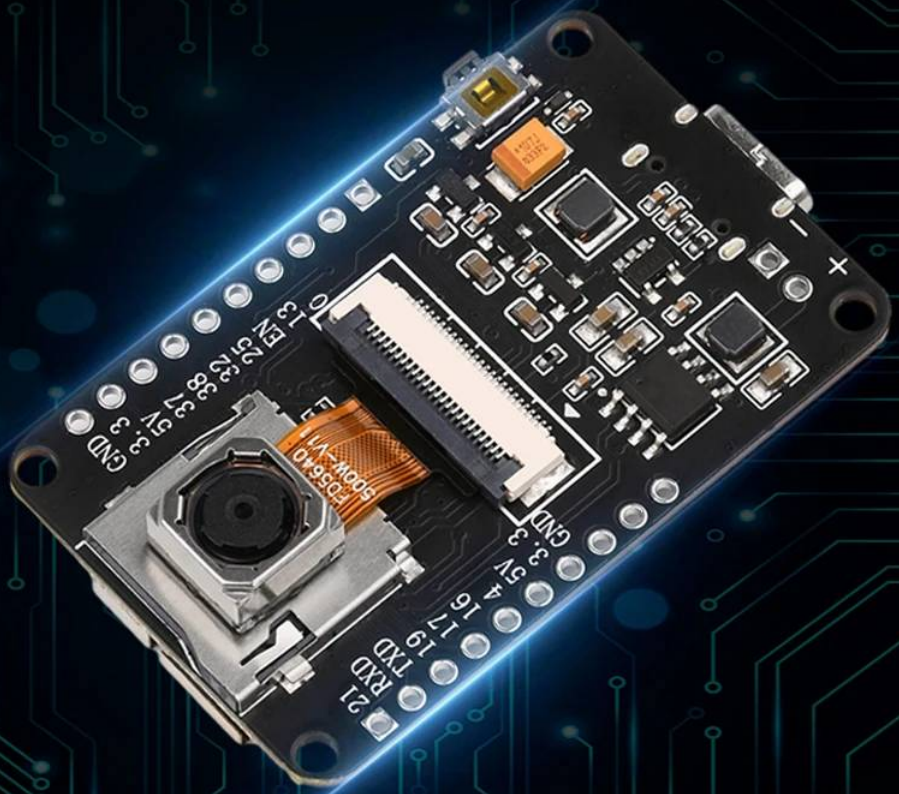
Integrated 2.4GHz Wi-Fi and BT dual mode

- ✔ Dual Power Mode
- ✔ Crisp, high-resolution graphics
- ✔ Support face recognition
- ✔ High performance & low power consumption



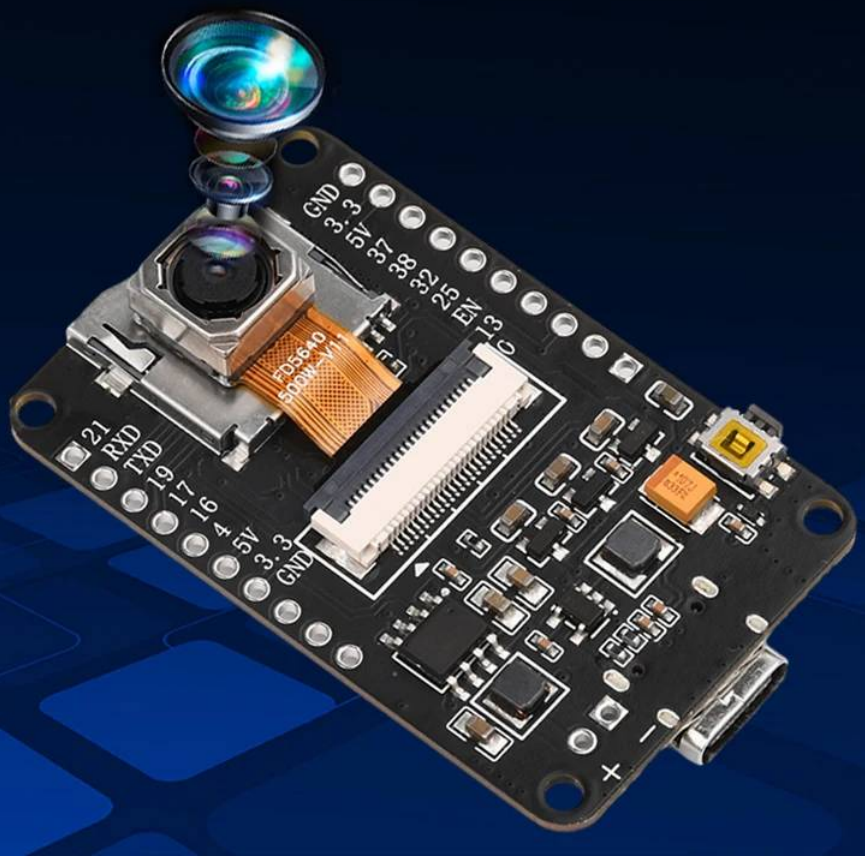
Low Power High Transmission

The module adopts ESP32 low power consumption program, integrated wifi and BT, the module can be operated independently as a minimum system.



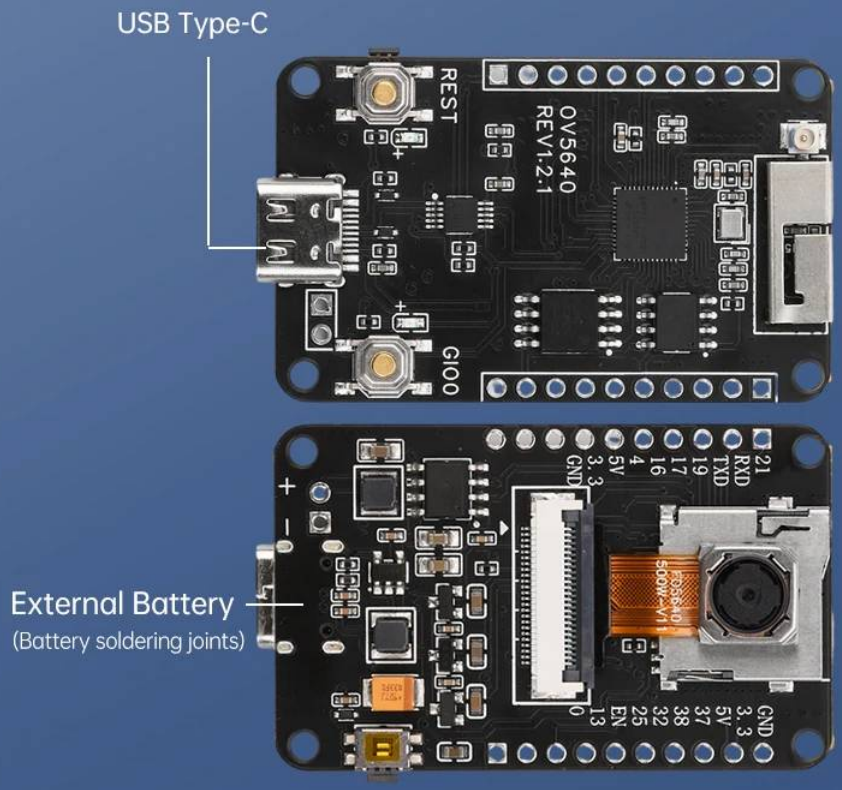
Powerful performance

The module is equipped with 0V5640 camera by default, with the powerful performance of ESP32, it can reach up to 30 FPS in VGA format.



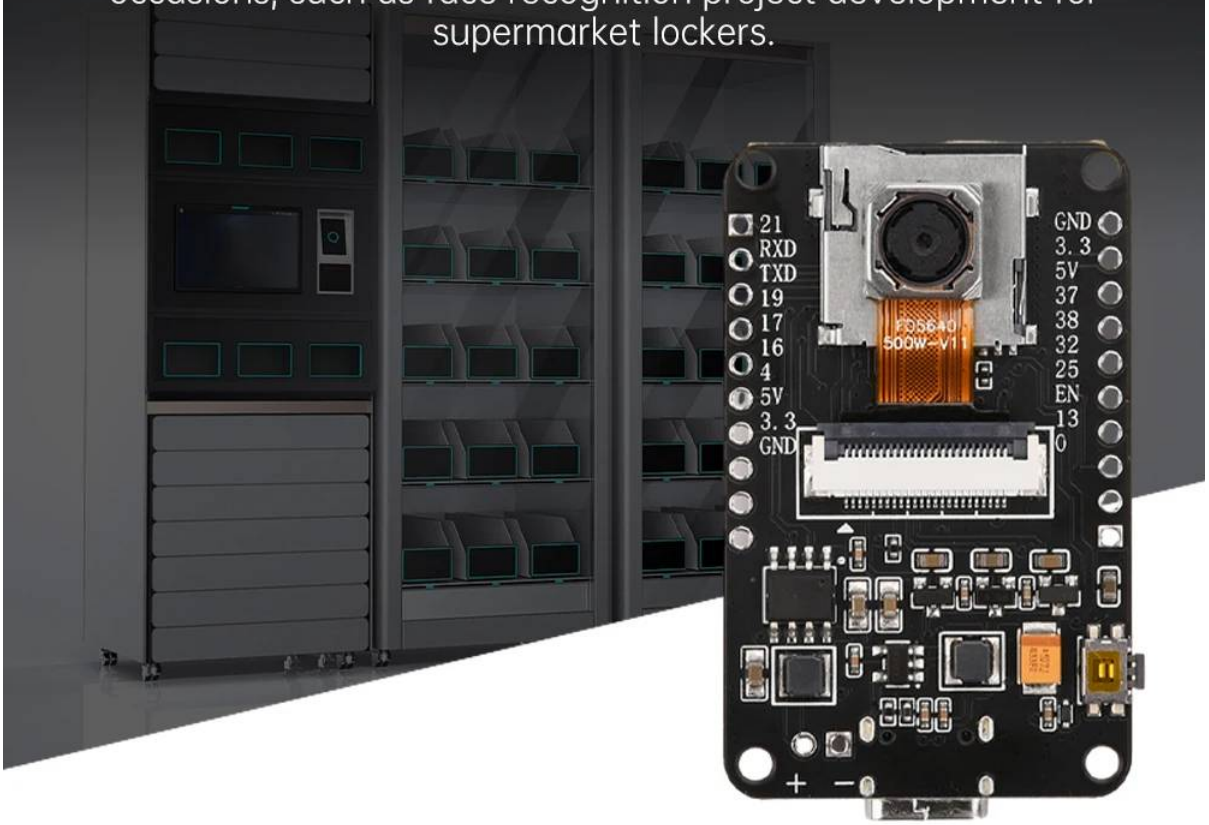
Fulfills a wide range of needs

The most notable features of this upgrade are the addition of 8M (bytes) of external RAM and an upgraded TF card slot, which allows users to use up to a 32G memory card for higher pixel counts and more image storage.



Support face recognition

And after joining PSRAM, it can also support face recognition framework ESP-WHO, which is suitable for IoT and smart device occasions, such as face recognition project development for supermarket lockers.



Product Features

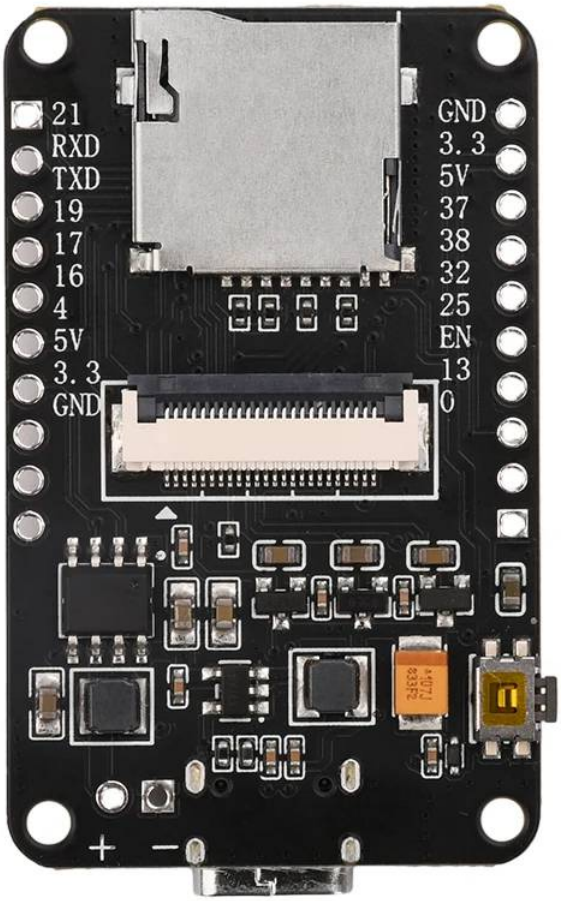
1, this development board is an upgraded version of the ESP32-CAM development board, increased PSRAM.

2, the module adopts the ESP32 low-power, program, integrated wifi and BT, the module can be used as the smallest system to run independently.

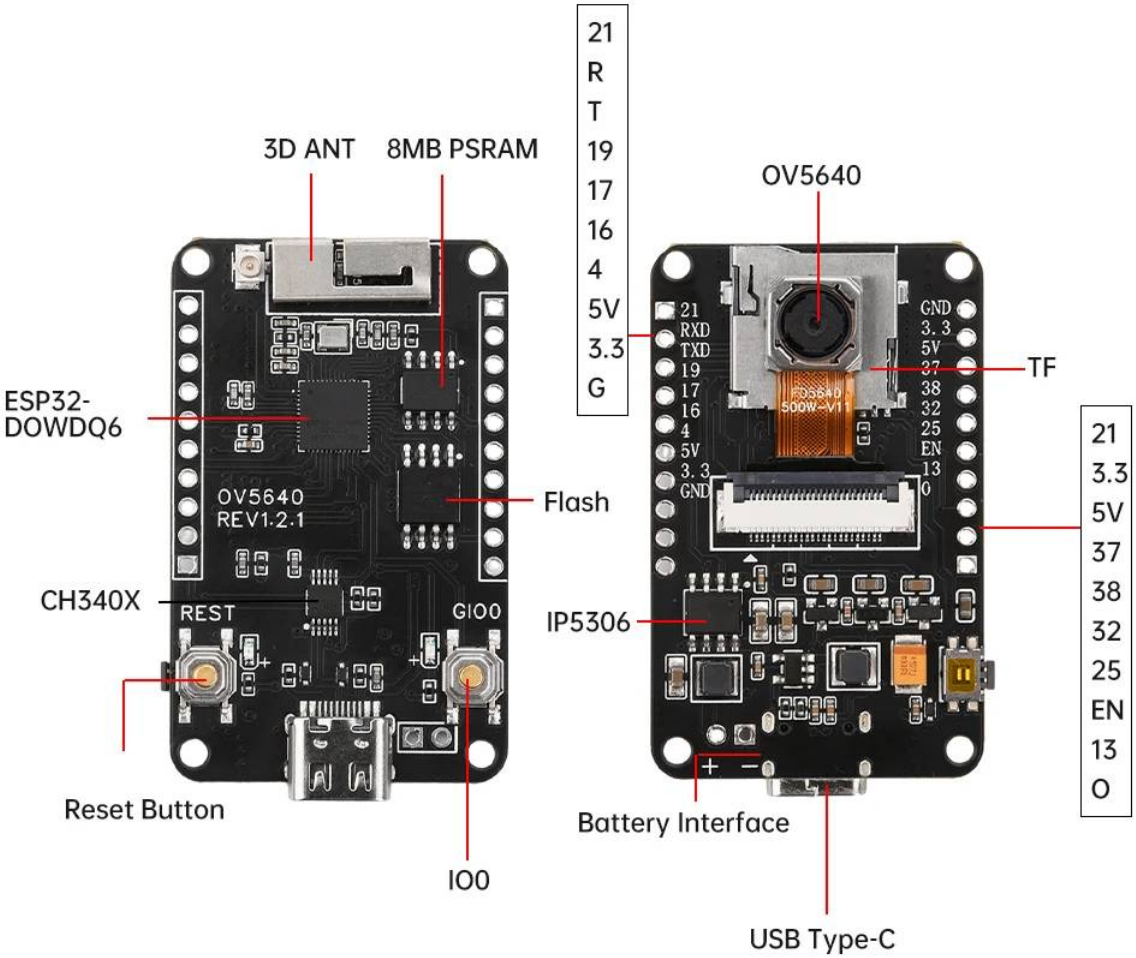
3, the module is equipped with the 0V5640 camera by default, with the strong performance of the ESP32, in the format of VGA up to 30 FPS.

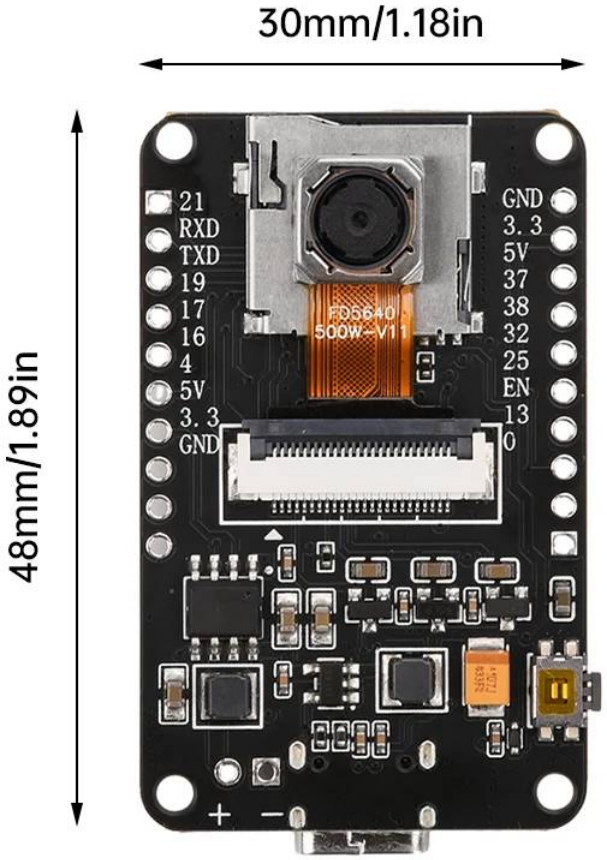
4, this time the The most significant feature of the upgrade is the addition of 8M (bytes) of external RAM and upgraded configuration of the TF card slot, which can support users to use a maximum of 32G memory card, which can be used to shoot a higher pixel and store more pictures, to meet the user's needs for a variety of secondary development.

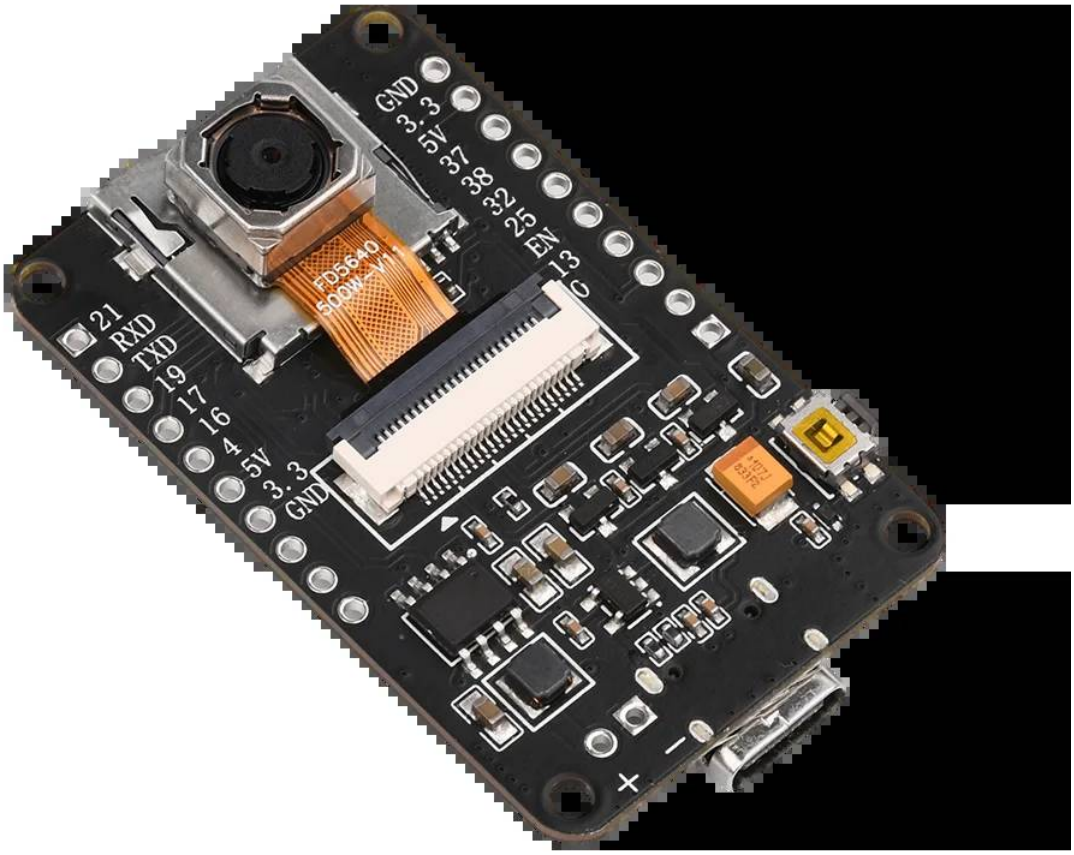
5, and after the addition of the PSRAM can also support the face recognition framework ESP-WHO, for the Internet of Things and intelligent equipment occasions, such as the face recognition of supermarket lockers. Supermarket locker face recognition project development.

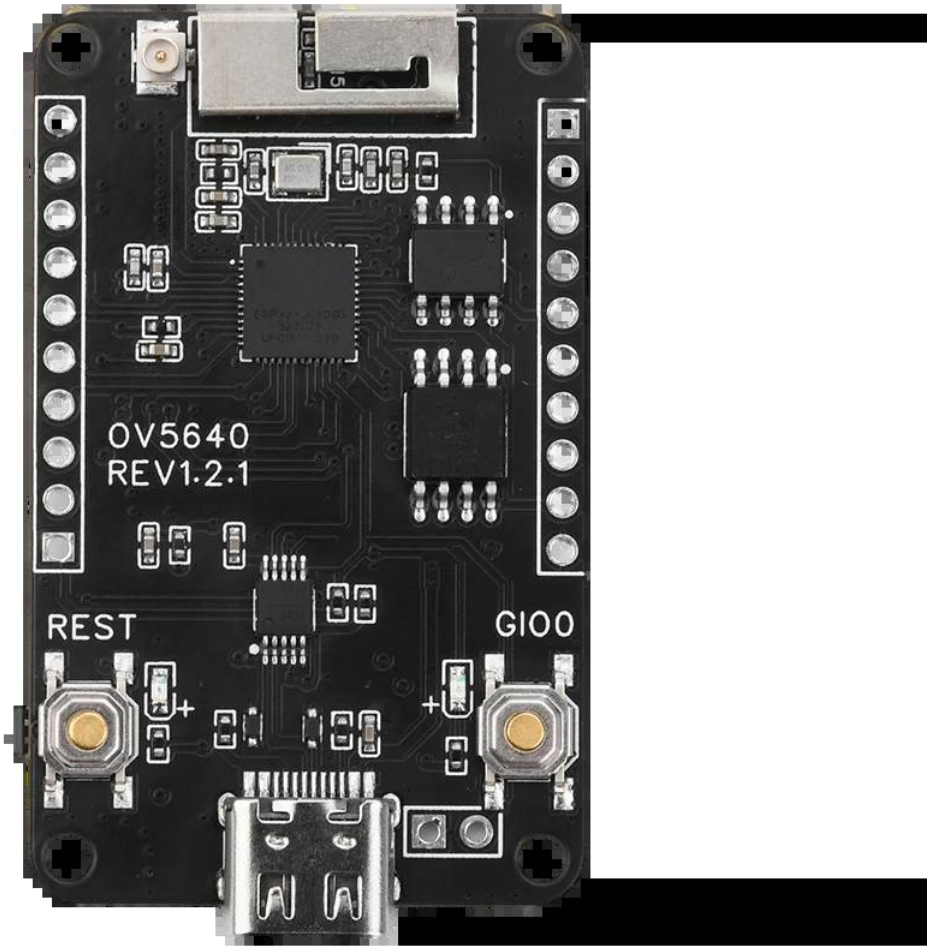

21
RXD
TXD
19
17
16
4
5V
3.3
GND
GND
3.3
5V
37
38
32
25
EN
13
0
+
−

FD5640
500W-V1.1


3D ANT
8MB PSRAM
ESP32-
DOWDQ6
OV5640
REV1.2.1
Flash
CH340X
REST
GIO0
Reset Button
IO0
21
R
T
19
17
16
4
5V
3.3
G
OV5640
TF
IP5306
Battery Interface
USB Type-C
21
3.3
5V
37
38
32
25
EN
13
O


30mm/1.18in
48mm/1.89in
21
RXD
TXD
19
17
16
4
5V
3.3
GND
GND
3.3
5V
37
38
32
25
EN
13
0
FD5640
500W-V11
+
−


21
RXD
TXD
19
17
16
4
5V
3.3
GND
GND
3.3
5V
37
38
32
25
EN
13
G
FD5640
SC00W_V1.1
+
-


OV5640
REV1.2.1
REST
GIO0